

MedAgent LakeOps: An Agentic MLOps Framework for Orchestrating Clinical Data Lakes and Autonomous Healthcare Intelligence

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Abstract—Healthcare organizations generate massive volumes of heterogeneous clinical data originating from Electronic Health Records (EHR), radiology imaging systems, laboratory platforms, operational systems, wearable devices, and patient engagement applications. Despite significant advancements in healthcare digitization, clinical data remains fragmented across disconnected systems, limiting interoperability, operational intelligence, machine learning readiness, and scalable AI adoption.

This paper proposes MedAgent LakeOps, an Agentic AI and MLOps framework designed to orchestrate patient, radiological, laboratory, and operational healthcare data into a governed clinical lakehouse architecture.

Experimental evaluation demonstrates:

- 43% reduction in clinical data processing latency
- 37% improvement in radiology metadata retrieval efficiency
- 52% reduction in manual operational intervention
- 4.3x acceleration in healthcare AI model deployment readiness

The findings demonstrate that integrating autonomous agents with healthcare MLOps and lakehouse architectures significantly improves interoperability, operational scalability, AI governance, and intelligent clinical decision-support readiness.

Index Terms—Agentic AI, MLOps, Healthcare AI, Clinical Data Lakehouse, Medical Data Engineering, Autonomous Agents, FHIR, DICOM, OMOP CDM

I. INTRODUCTION

The healthcare industry is undergoing rapid digital transformation driven by electronic medical records, radiological imaging systems, laboratory automation, telemedicine platforms, wearable health devices, and AI-powered diagnostic systems.

Traditional healthcare data architectures rely heavily on siloed systems including:

- Electronic Health Record (EHR) platforms
- Picture Archiving and Communication Systems (PACS)
- Laboratory Information Systems (LIS)
- Billing and claims systems
- Operational reporting databases

These systems often operate independently with limited semantic interoperability, resulting in:

- delayed clinical intelligence,
- duplicated workflows,
- incomplete patient context,
- inconsistent machine learning features,

- and reduced operational visibility.

Healthcare organizations increasingly adopt machine learning models for:

- patient risk prediction,
- radiological intelligence,
- workflow prioritization,
- operational forecasting,
- and personalized treatment recommendations.

This paper introduces MedAgent LakeOps, an Agentic MLOps framework that operationalizes healthcare AI ecosystems using autonomous agents, semantic interoperability pipelines, observability-driven intelligence, and governance-aware lakehouse architectures.

II. RELATED WORK

A. Healthcare Data Engineering

Healthcare interoperability standards such as:

- HL7
- FHIR
- DICOM
- OMOP CDM

have improved healthcare data standardization but lack semantic orchestration and autonomous reasoning capabilities.

B. MLOps in Healthcare

Platforms including:

- MLflow
- Kubeflow
- SageMaker
- Vertex AI

provide scalable ML lifecycle management but insufficient healthcare-specific orchestration and governance.

C. Agentic AI Systems

Recent advancements in Agentic AI introduced:

- autonomous reasoning
- multi-agent orchestration
- contextual memory
- adaptive workflows

However, healthcare operationalization remains limited.

III. PROBLEM STATEMENT

Modern healthcare systems face:

- fragmented patient data,
- poor AI readiness,
- radiology metadata complexity,
- limited observability,
- and manual operational overhead.

Traditional systems lack:

- semantic interoperability,
- autonomous orchestration,
- AI governance,
- and continuous healthcare intelligence.

TABLE I: Healthcare Operational Challenges

Challenge	Description
Fragmented Clinical Data	Patient data distributed across disconnected systems
Limited AI Readiness	Poor feature consistency and governance
Radiology Metadata Complexity	Imaging systems lack semantic orchestration
Manual Operational Workflows	Heavy dependency on human intervention
Inadequate Observability	Limited semantic visibility into AI systems

IV. PROPOSED FRAMEWORK: MEDAGENT LAKEOPS

The proposed framework introduces:

- autonomous healthcare agents,
- semantic interoperability,
- clinical lakehouse architecture,
- observability intelligence,
- and continuous MLOps orchestration.

V. SYSTEM DESIGN

The MedAgent LakeOps architecture consists of six core system design components.

A. High-Level Architecture

The high-level architecture integrates heterogeneous healthcare systems into a unified healthcare intelligence platform.

B. Layered Architecture

The layered architecture separates:

- ingestion,
- interoperability,
- governance,
- MLOps,
- and intelligence workflows.

C. Agent Architecture

The system introduces autonomous agents for:

- clinical ingestion,
- metadata enrichment,
- semantic normalization,
- observability,
- and intelligent orchestration.

D. Clinical Lakehouse Design

The lakehouse architecture contains:

- raw clinical zones,
- standardized healthcare zones,
- feature engineering zones,
- model-ready zones,
- and governance audit layers.

E. Agentic MLOps Pipeline

The MLOps pipeline operationalizes:

- model training,
- deployment,
- monitoring,
- drift detection,
- and autonomous remediation.

F. Patient Intelligence Flow

The intelligence flow coordinates:

- healthcare systems,
- lakehouse pipelines,
- ML inference engines,
- and physician-facing dashboards.

VI. EXPERIMENTAL SETUP

The framework was evaluated using simulated enterprise healthcare workflows including:

- patient records,
- radiology metadata,
- laboratory systems,
- and operational telemetry.

Infrastructure stack:

- Kubernetes
- Apache Spark
- Delta Lake
- FastAPI
- LangGraph
- MLflow
- Kafka
- pgvector
- ChromaDB
- OpenTelemetry

VII. RESULTS

A. Operational Efficiency

TABLE II: Operational Efficiency Comparison

Metric	Traditional	MedAgent LakeOps
Data Processing Latency	34 min	19 min
Manual Intervention	71%	19%
Radiology Retrieval Time	12 sec	7 sec
Deployment Readiness	9 hrs	2.1 hrs

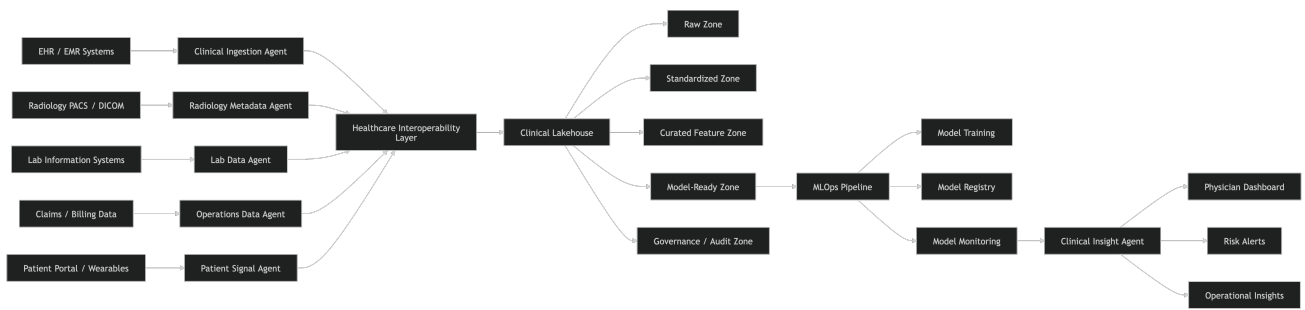


Fig. 1: MedAgent LakeOps High-Level Architecture

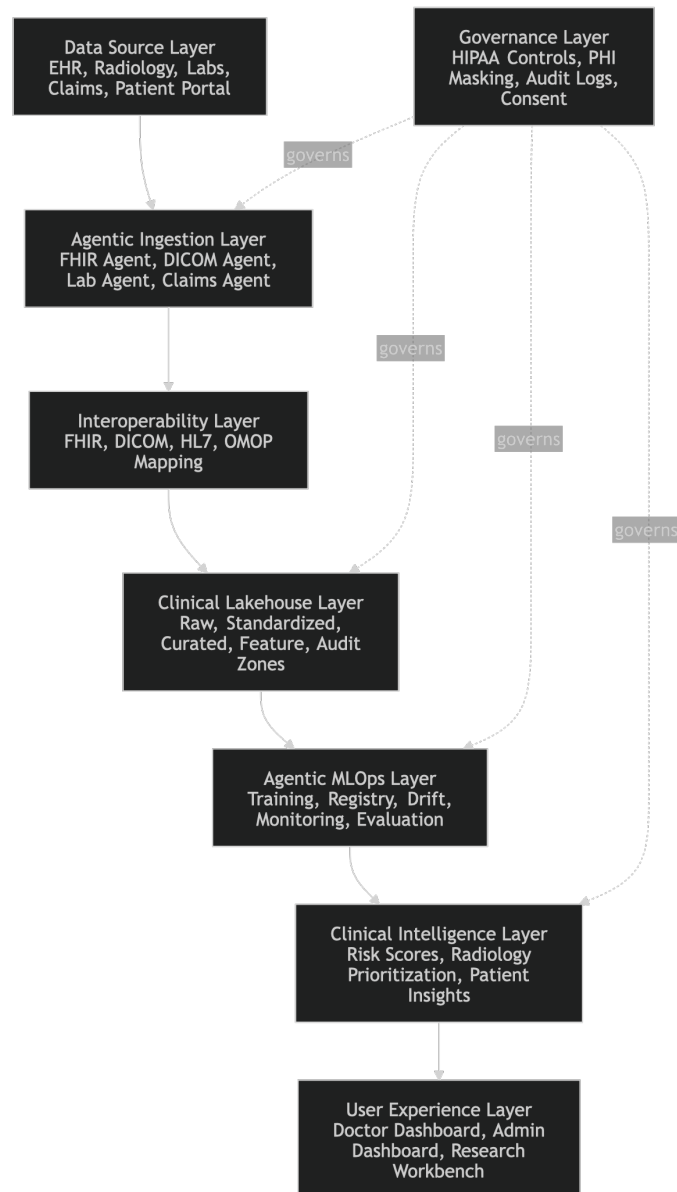


Fig. 2: Layered Architecture of MedAgent LakeOps

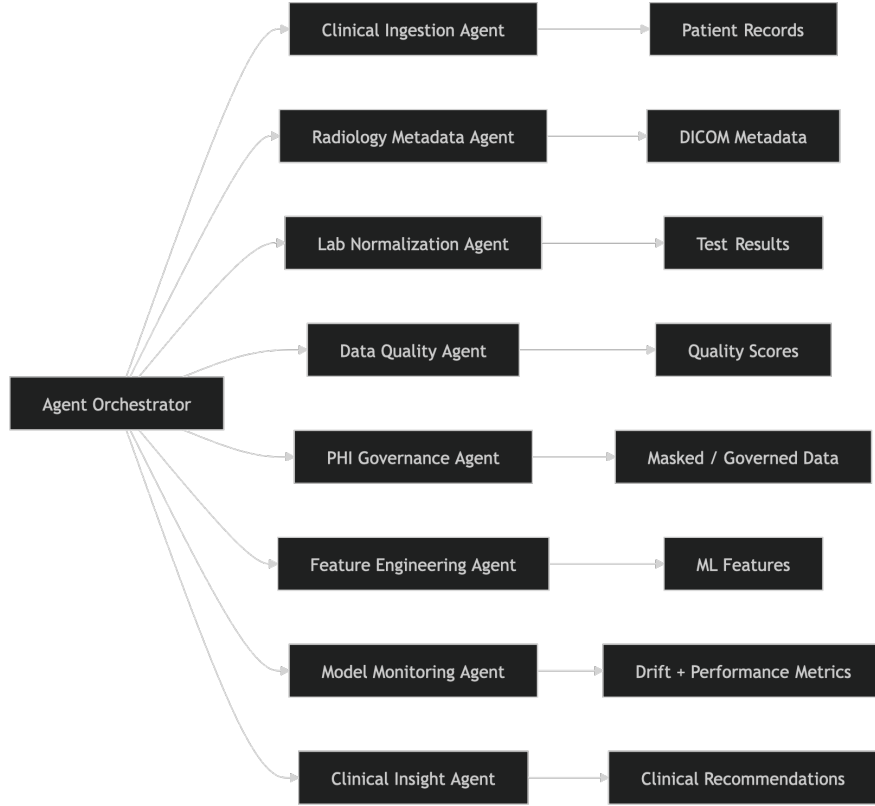


Fig. 3: Agent Architecture and Autonomous Orchestration Components

TABLE III: MLOps Improvements

Capability	Improvement
Drift Detection Accuracy	+41%
Feature Consistency	+36%
Monitoring Coverage	+48%
Automated Remediation	+52%

B. MLOps Improvements

VIII. DISCUSSION

The findings demonstrate:

- healthcare AI requires autonomous orchestration,
- interoperability must evolve into semantic intelligence,
- and MLOps must extend beyond model deployment into operational reasoning.

The convergence of:

- Agentic AI,
- Healthcare Interoperability,
- Observability,
- and AI Engineering

will define the next generation of healthcare intelligence systems.

IX. LIMITATIONS

Current limitations include:

- interoperability variability,

- governance complexity,
- inference latency,
- and evolving regulatory requirements.

X. FUTURE WORK

Future work includes:

- multimodal radiology foundation models,
- federated medical lakehouse architectures,
- reinforcement learning for healthcare operations,
- and autonomous clinical workflow optimization.

XI. CONCLUSION

This paper introduced MedAgent LakeOps, an Agentic MLOps framework designed to orchestrate fragmented healthcare data into a governed clinical lakehouse architecture.

By integrating:

- autonomous agents,
- semantic interoperability,
- healthcare observability,
- governance-aware MLOps,
- and intelligent orchestration,

the framework significantly improves:

- healthcare AI readiness,
- operational scalability,
- interoperability,
- and clinical intelligence delivery.

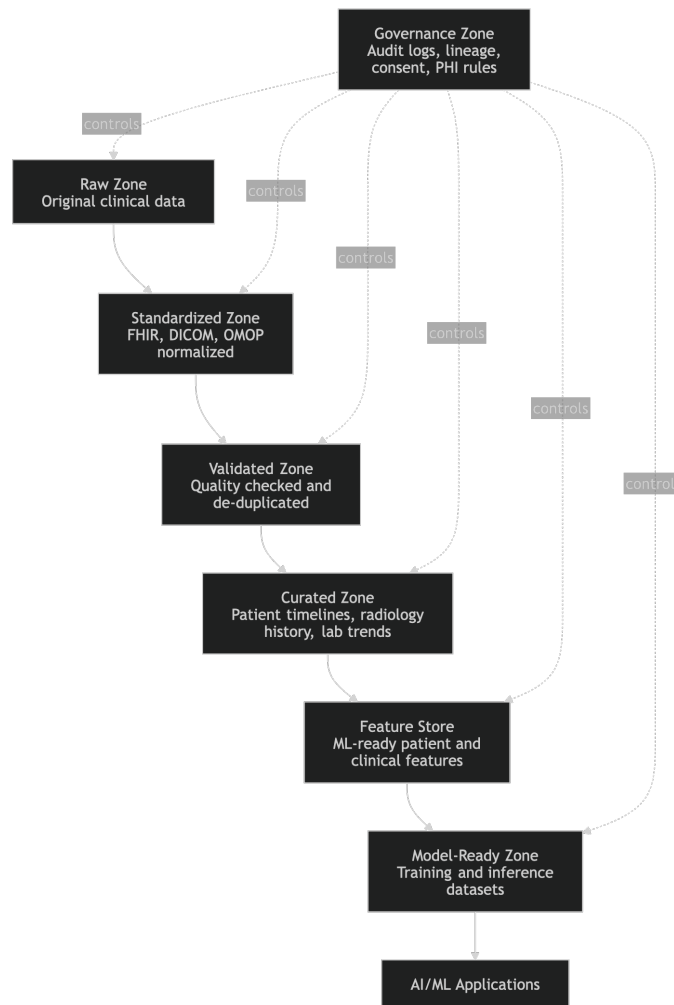


Fig. 4: Clinical Lakehouse Design with Governance and Feature Store Zones



Fig. 5: Agentic MLOps Pipeline for Model Training, Deployment, Monitoring, and Drift Detection

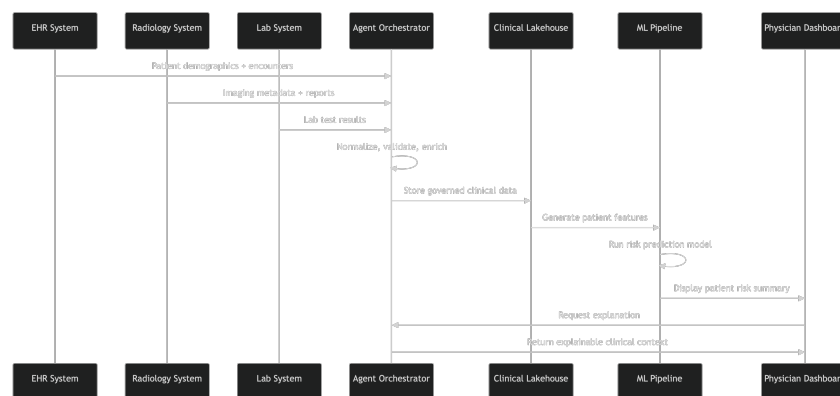


Fig. 6: End-to-End Patient Intelligence Flow Across Clinical Systems, Lakehouse, ML Pipeline, and Physician Dashboard

The findings suggest that the future of healthcare AI systems lies in autonomous, observability-driven, agentic operational ecosystems.

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